Group 1 · Ship Design Project 2026

11 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

How the work is split. The in-class exercises are individual practice of each method on the reference ship (the 37,600 t oil/chemical tanker) and are not graded. This project is your group's work, done outside class: you apply each method to the Azores Feeder and complete the full scope of every deliverable below. A report that only repeats the class exercises does not meet the scope.

Common deliverables D1–D10

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D1	Design Basis	Ex. 1.2 does the owner's brief add up, classify the ship · Ex. 1.3 design phases and documents	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked
D2	Reference Ship Database	Ex. 1.4 first Δ, L, B, T, D from the App. A regressions, with 95% bands	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds
D3	Ship Model	Ex. 1.5 displacement equation · Ex. 2.3 main dimensions · Ex. 2.4 power · Ex. 3.1 weights · Ex. 3.2 capacity · Ex. 3.3 engine and propeller	Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance
D4	Voyage and Financial Model	Ex. 3.4 round voyage and fuel · Ex. 4.1 building cost · Ex. 4.2 RFR and NPV	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input
D5	Validation Note	No class exercise: done by the group	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed
D6	Optimization Report	Ex. 5.1 synthesis model · Ex. 5.2 optimization (demonstration + individual step) · Ex. 5.3 multi-objective trade-off	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D7	Compliance Sheet	Ex. 4.3 GM, freeboard and attained EEDI	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port
D8	General Arrangement and Hull Form	Ex. 5.4 sectional area curve and LCB · Ex. 6.1 sections, bow and stern · Ex. 6.2 GA and capacity plan	Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · hydrostatics check
D9	Specialization Documents (2)	No class exercise: self-study decks	One calculation note per specialization, with its required drawings (see Your specializations)
D10	Datasheet and Presentation	No class exercise	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.